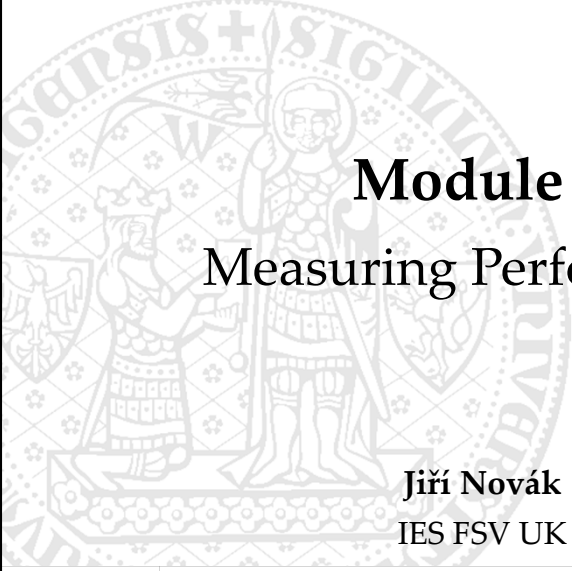




Module 7

Measuring Performance

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Recap



- **Contents**
 - summarizes cash *receipts* and cash *outlays* that occur during fiscal year
- **Classification**
 - classified into *operating*, *investing*, and *financing*

Statement of Cash Flow

cash flow from **operating** activities

+/- cash flow from **investing** activities

+/- cash flow from **financing** activities

total cash flow



Recap

▪ **Cash Flow**

$$\Delta \text{cash} = \Delta E_q + \Delta Li - \Delta \text{other As}$$

$$\Delta \text{cash} = \underbrace{\Delta CC + NI - Dv}_{\text{... CFO}} + \underbrace{\Delta COL + \Delta CFL + \Delta LTFL}_{\text{... CFI}} - \underbrace{\Delta COA - \Delta CFA - CapEx + Dp}_{\text{... CFF}}$$

$$\Delta \text{cash} = NI + Dp - \Delta COA + \Delta COL - \dots \text{CFO}$$

$$- CapEx + \dots \text{CFI}$$

$$+ \Delta CC - Dv + \Delta LTFL + \Delta CFL - \Delta CFA \dots \text{CFF}$$

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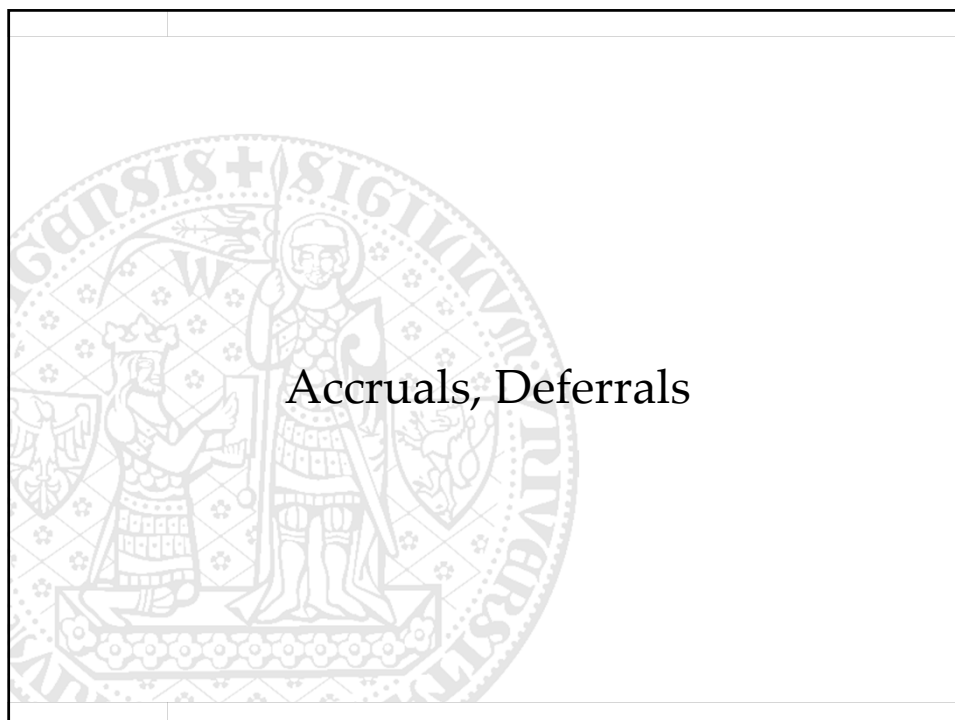
Recap



Indirect Statement of Cash Flow	
	net income
+	depreciation
-	increase in current assets
+	increase in current operating liabilities
cash flow from operations	
-	capital expenditures
-	financial investments
cash flow from investing	
+(-)	issuance (retirement) of debt
+(-)	issuance (repurchase) of common stock
-	dividends paid out
cash flow from financing	



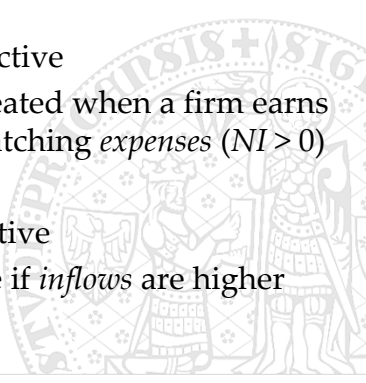
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Timing Discrepancies

- **Purpose**
 - both income and cash flow reflect *performance*, but they measure it differently
- **income (IS)**
 - *value* generation perspective
 - value assumed to be created when a firm earns higher *revenues* than matching *expenses* ($NI > 0$)
- **cash flow (SCF)**
 - *cash* generation perspective
 - net cash flow is positive if *inflows* are higher than *outflows*



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Timing Discrepancies

Relationship

$$\sum_{t=1}^{\infty} NI_t = \sum_{t=1}^{\infty} CF_t$$

- over entire firm existence sum of net *income* is equal to sum of net *cash flows*, but net income is *not* equal to net cash flows in every fiscal period
- if value is created but cash is not collected (or vice versa) value has to be *stored* on BS till the time it is converted into cash
- BS is a storage of value
 - *assets* store *positive* value (future benefit)
 - *liabilities* show *negative* value (reduct. benefit)

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Timing Discrepancies



Timing Sequence

- What is *most common* timing sequence of positive and negative income and cash flows related to *operating* activities?

	negative	positive
cash	outflow	inflow
income	expense	revenue

- Different timing of cash outflows and inflows implies that firms need ... ?



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Timing Discrepancies



▪ Accruals & Deferrals

- discrepancies in timing of income & cash flows *temporarily* recorded as assets & liabilities on BS
- when timing discrepancies are *closed*, effect on assets & liabilities is eliminated

	expenses	revenues
deferrals (cash first)	pre-paid expense (asset)	unearned revenue (liability)
accruals (cash later)	accrued expense (liability)	accrued revenue (asset)

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Timing Discrepancies

▪ Accrued Revenue

- *revenue* recognition (positive income) before *inflow* (positive cash flow)
- *accounts receivable* (As) temporarily store value that will later be converted into cash (benefit)

Assets		Liabilities, Equity	Expenses	Revenues
Accounts Receivable				Net Sales
① 300	② 300			① 300
Cash				
② 300				

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Timing Discrepancies

▪ Accrued Expense

- *expense* accrue (negative income) before *outflow* (negative cash flow)
 - *salaries payable* (Li) temporarily store negative value, cash will be paid to employees (reduction in future benefit)

Assets		Liabilities, Equity		Expenses	Revenues
Cash		Accrued Expense		Expense	
	② 150	② 150	① 150	① 150	

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Timing Discrepancies

▪ Unearned Revenue

- *inflow* (positive cash flow) before *revenue* recognition (positive income)
 - *unearned revenue* (Li) temporarily store negative value, goods or services will be provided (reduction in future benefit)

Assets		Liabilities, Equity		Expenses	Revenues
Cash		Unearned Revenue			Net Sales
① 200		② 200	① 200		② 200

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Timing Discrepancies

▪ Deferred Expenses

- *outflow* (negative cash flow) before accrued *expenses* (negative income)
 - *inventories, pre-paid expenses (As)* temporarily store value that will later be converted into cash or used (future benefit)

Assets		Liabilities, Equity	Expenses	Revenues
Cash				
	① 100			
Prepaid Expense			Expense	
① 100			② 100	
	② 100			

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Timing Discrepancies



▪ Storage of Value

- assets temporarily store *positive value*
 - value generated (*Rv*), cash not yet received (*AR*)
 - cash paid but benefit not consumed (*lv*)
- liabilities temporarily store *negative value*
 - benefit consumed (*Ex*), cash not yet paid (*AP*)
 - cash received, but value not yet generated (*UR*)

▪ Closing

- *temporary storage (As, Li)* reduced to zero when timing discrepancy between *NI* & *CF* is closed

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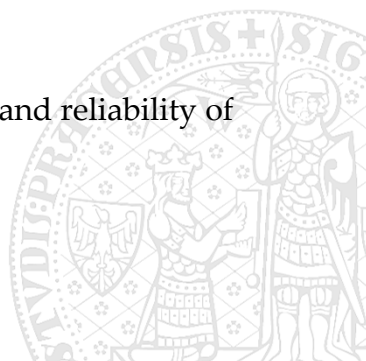


Questions

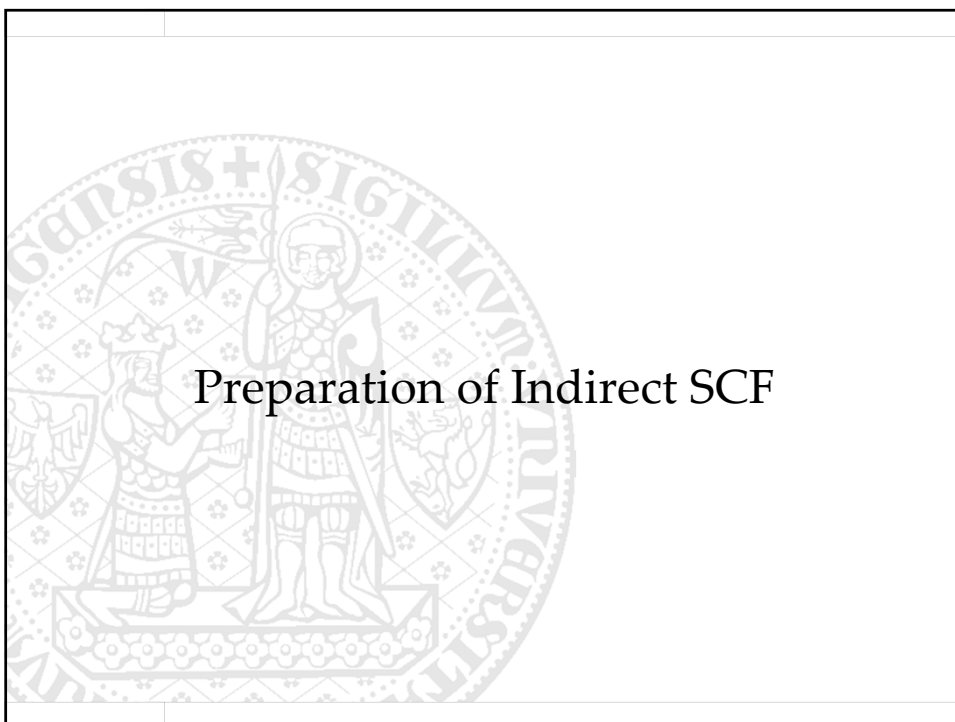


- **Sections of IS and SCF**
 - Why does the Statement of Cash Flow have three sections while the Income Statement just two?
- **Relevance vs. Reliability**
 - Assess relative relevance and reliability of earnings and cash flows?

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Preparation of Indirect SCF





Statement of Cash Flow

- **Reversing Accruals & Deferrals**
 - all timing discrepancies between income & cash flow must be reflected (temporarily stored) as *changes* in assets & liabilities on BS
 - one may use BS changes to *compute* cash flows from income by reversing accrual adjustments

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Statement of Cash Flow

- **Methods**
 - CFO can be prepared either by *direct* or *indirect* method; both methods produce the same CFO
 - CFI, CFF are always prepared by direct method
- **Preparation**
 1. consider income from IS income as:
 - *lump sum* (indirect method) *or*
 - *item by item* (direct method)
 2. calculate *changes* of BS accounts (accruals), decide if they are cash *inflows* or *outflows*
 3. categorize as *operating, financing, investing*

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Statement of Cash Flow

- **Indirect Method**

- required by US GAAP, IFRS
- BS shows storage of value *levels* at fiscal year end
- indirect SCF shows *changes* in individual balance sheet accounts over fiscal year end

Indirect Method	
<i>NI</i>	
+ <i>Dp</i>	depreciation and non-cash expenses
– ΔCOA	increases in current operating assets
+ ΔCOL	increases in current operating liability.
<i>CFO</i>	

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Indirect Method



- **Non-Cash Expenses**

- *depreciation*
 - expense that does *not* involve any current cash outlay
 - depreciation is subtracted in determining *NI*, it has to be added back to obtain *CFO*

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Indirect Method

- **Prevent Double Counting**

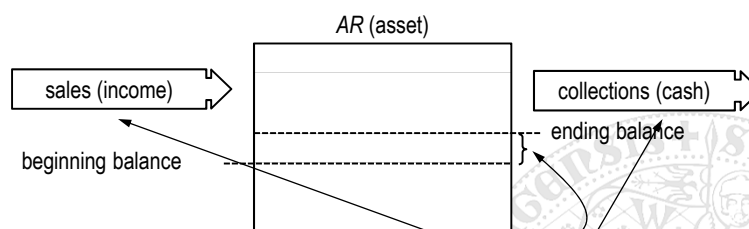
- *gains on sale of fixed assets*
 - entire cash proceeds are included in *CFI*
 - gains must be subtracted from *NI* not to double count them in both *CFI* and *CFO*

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Indirect Method

- **Accounts Receivable**



- **increase in AR**
 - ending > beginning → cash inflow < revenue (–)
 - cash collections from customers were less than IS revenues (Δ must be subtracted from *NI*)

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Indirect Method

▪ Reversal of Accrual Adjustments

- accrual adjustments are discrepancies between revenues/expenses and corresponding cash flows, e.g. sales and cash collections from customers
- they are reflected in changes in current assets and current liabilities on BS

▪ Corresponding Accounts

current asset	revenue/expense	current liability
AR	sales	unearned revenue
inventory	COGS	AP
prepaid expense	operating expense	accrued liability

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Indirect Method

Indirect Statement of Cash Flow

net income
+ depreciation
– increase in current assets
+ increase in current operating liabilities
cash flow from operations
– capital expenditures
– financial investments
cash flow from investing
+(-) issuance (retirement) of debt
+(-) issuance (repurchase) of common stock
– dividends paid out
cash flow from financing

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Questions

- **Cash Paid to Suppliers**

- One of the examples used in the lecture shows how to find the amount of cash paid to suppliers of raw materials in case the firm pays for inventories immediately in cash.
- Relaxing the assumption of all inventories are purchased for cash, how can we determine the cash paid for inventory if we only have the Income Statement and the Balance Sheet?

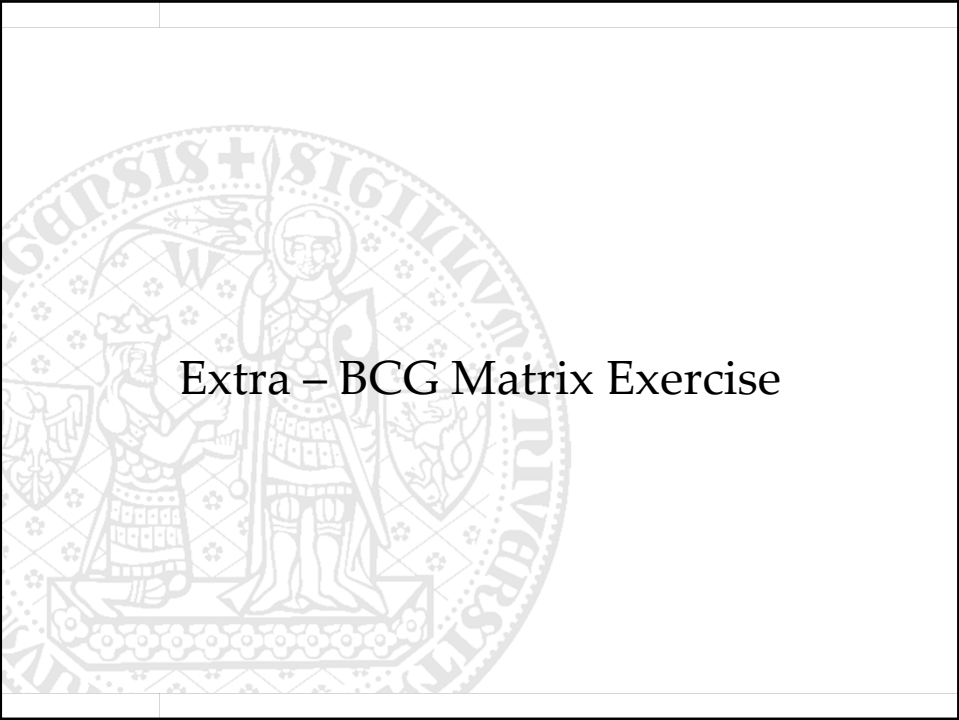
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Cash Flow



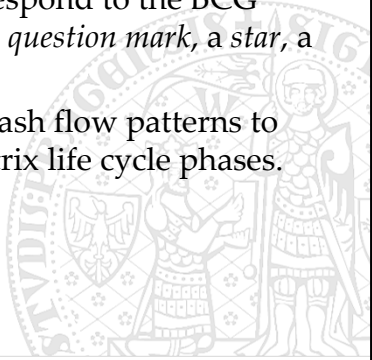
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Extra – BCG Matrix Exercise



Exercise

- **CF over Life Cycle**
 - Slides below show cash flows of four firms in different phases of their life cycle (A, B, C, D).
 - The life cycle phases correspond to the BCG classification matrix, i.e. a *question mark*, a *star*, a *cash cow*, a *lousy dog*.
 - Attribute the individual cash flow patterns to the different the BCG matrix life cycle phases.
- 



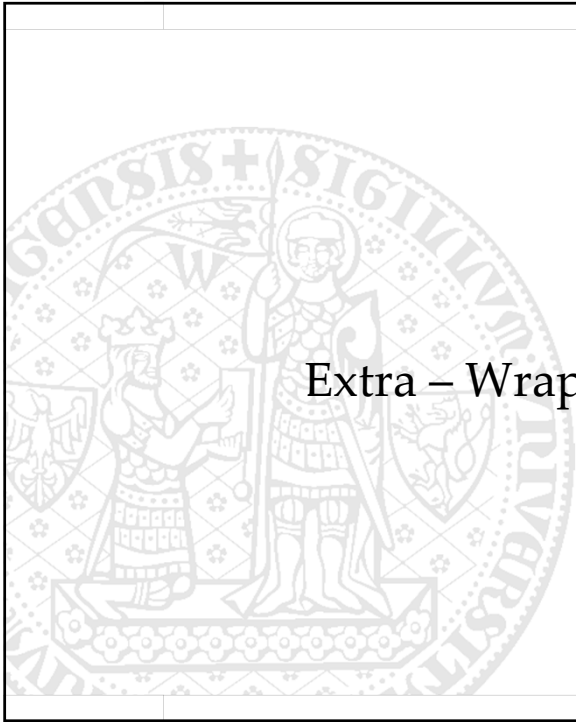
Exercise

▪ BCG Matrix

	A	B	C	D
CFO	(3)	7	15	8
CFI	(15)	(12)	(8)	2
CFF	18	5	(7)	(10)
net CF	0	0	0	0



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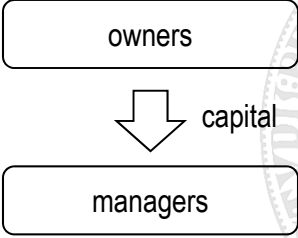
Extra – Wrap Up



Accounting



- **Importance**
 - accounting facilitates investment as commitment to *regular* reporting independently *audited* helps to build the necessary trust between:
 - *investors* (owners, creditors) who have capital
 - *managers* who have competence to use it



```
graph TD; A[owners] -- capital --> B[managers]
```

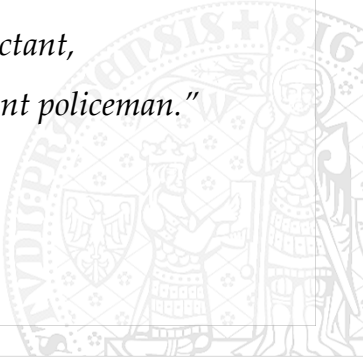
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Accounting

Transparency

*"Sunlight is the best disinfectant,
electric light the most efficient policeman."*



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Accounting



Transparency

“Accounting does not make corporate earnings or balance sheets more volatile. Accounting just increases the transparency of volatility in earnings.”

Diane Garnick

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Admin

- **Company Valuation (JEM132)**
 - use accounting information to estimate *intrinsic value* of firms
 - valuable for equity *investment* decisions (“value investing”)
 - put your knowledge into *use*
- **Bachelor’s or Master’s Theses Supervision**
 - good *interaction* in the class, perhaps interest in accounting, analysis, and finance topics
 - potentially form a *mini-group*?

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Admin

- **Course Evaluation**
 - in business *accounting* is used to evaluate performance of managers
 - universities use *course evaluation* to assess performance of teachers
 - please, *fill in* the evaluation and be as *specific* as possible to let me improve the course

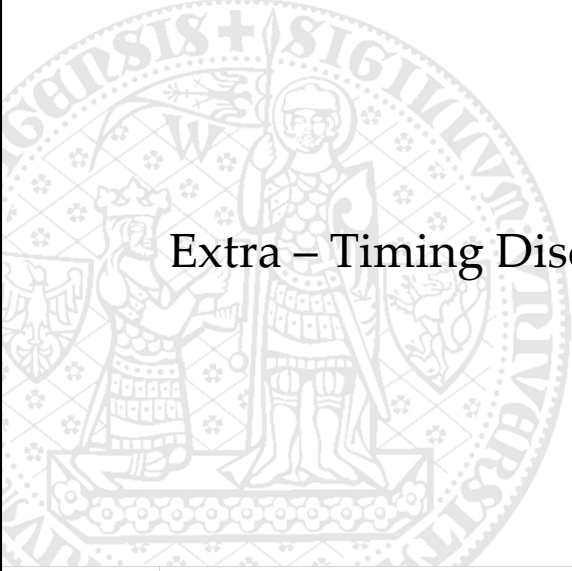
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Thank You!



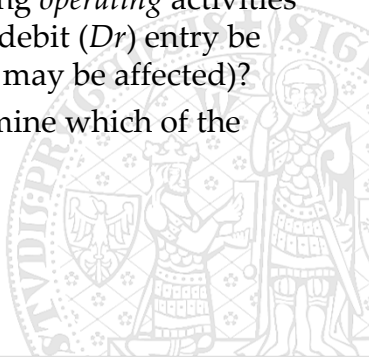
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Extra – Timing Discrepancies



Timing Discrepancies

- **Cash Payments**
 - Payments of cash is recorded as credit (*Cr*) entries at cash account on BS.
 - When firm *pays* cash during *operating* activities where may the offsetting debit (*Dr*) entry be (i.e. what kind of account may be affected)?
 - What criterion will determine which of the alternatives will be used?
- 



Timing Discrepancies

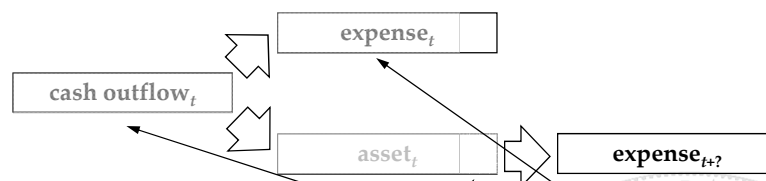
▪ Cash Payments

- cash paid for services is classified based on timing of corresponding economic benefit
 - *Ex* – contemporary economic benefit (repair)
 - $\uparrow$ *As* – future economic benefit (insurance)
 - $\downarrow$ *Li* – past economic benefit (salaries)
- cash paid for inventory temporarily “stored” as *asset*, becomes *expense* when delivered to clients

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Timing Discrepancies



- firm purchased for cash €100 of inventory (merchandise) and sold inventory worth €80

COGS	- €80
- ΔIv	- (+ €20)
cash payments	- €100

... subtract an increase in *As*

- cash outflow of €20 has not yet reached IS, cash outflow was larger than expenses in IS

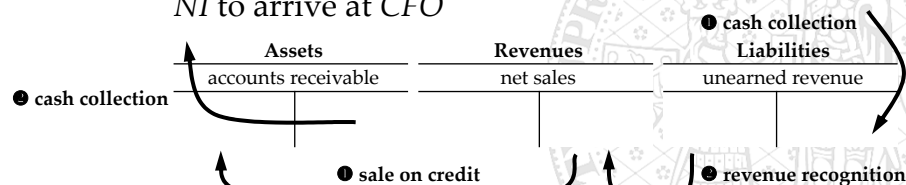
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Extra – Corresponding Accounts

Indirect Method

▪ Sales → Cash Collections

- $\uparrow$ AR means that sales revenue $>$ cash collections;
 Δ AR reflects non-cash sales that should be subtracted from sales in *NI* to obtain *CFO*
- $\uparrow$ unearned revenue: sales $<$ cash collections;
 Δ unearned revenue represents cash collected in excess of sales revenue that should be added to *NI* to arrive at *CFO*





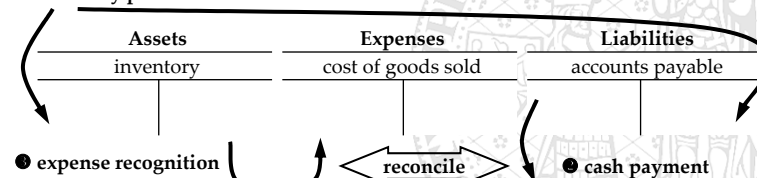
Indirect Method

▪ Cost of Goods Sold

- COGS are subtracted from sales in determining *NI*; if purchases > COGS $\Rightarrow$ $\uparrow$ inventories; if inventory purchases exceed COGS the increase in inventories must be subtracted from *NI* to get *CFO*

- $\uparrow$ AP: inventory purchases > cash payments; Δ AP should be added to *NI* to get *CFO*

● inventory purchase



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Extra – Direct Statement of Cash Flow



Direct Method

- **Direct Method**
 - groups BS accruals with corresponding IS revenue or expense line item
 - because the two methods differ only in ordering of the items, they must yield the same *CFO*
 - closely mimics IS structure, it can be seen as a “cash flow version of IS,” useful to discern the origin of individual flows

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Direct Method

Direct Statement of Cash Flow	
cash collection from customers	~ net sales
– cash paid to suppliers	~ COGS
– cash paid to employees	} ~ SG&A
– cash paid for other operating expenses	
– cash paid for interest	~ fin. income
+ cash revenue from interest	} ~ fin. expense
+ cash revenue from dividends	
– cash paid for taxes	~ taxes
cash flow from operations	~ net income

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